



9594 - Planck Bullet: New Dark Matter Supercollider

Cycle: 5, Proposal Category: GO

INVESTIGATORS

<i>Name</i>	<i>Institution</i>
Gregor Rihtarsic (PI)	University of Ljubljana, Dept. of Physics
Prof. Marusa Bradac (CoI) (CoPI)	University of Ljubljana, Dept. of Physics
Dr. Chris J. Willott (CoI) (CSA Member)	NRC Herzberg Institute of Astrophysics
Prof. Annika Peter (CoI) (US Admin CoI)	The Ohio State University
Dr. Andrew Robertson (CoI)	Carnegie Institution of Washington
Dr. Stacy Kim (CoI)	Mount Holyoke College
Dr. Tim Schrabback (CoI) (ESA Member)	University of Innsbruck
Dr. Anishya Harshan (CoI) (ESA Member)	University of Cambridge, Kavli Institute for Cosmology
Dr. Brian Lemaux (CoI)	NOIRLab - Gemini North (HI)
Dr. Marcin Sawicki (CoI) (CSA Member) (CoPI)	Saint Mary's University
Dr. Nicholas Martis (CoI)	University of Ljubljana, Dept. of Physics
Dr. Yoshihisa Asada (CoI) (CSA Member)	University of Toronto
Guillaume Desprez (CoI) (ESA Member)	Kapteyn Astronomical Institute
Lamiya Mowla (CoI)	Wellesley College
Prof. Douglas Clowe (CoI)	Ohio University
Dr. Christine Jones (CoI)	Smithsonian Institution Astrophysical Observatory
Prof. Anthony Hernan Gonzalez (CoI)	University of Florida
Dr. Vladan Markov (CoI)	University of Ljubljana, Dept. of Physics
Jon Judez (CoI)	University of Ljubljana, Dept. of Physics
Dr. Roberta Tripodi (CoI) (ESA Member)	INAF - Osservatorio Astronomico di Roma
Dr. Gael Noirot (CoI) (CSA Member)	Space Telescope Science Institute - CSA - JWST
Vesna Pirc Jevsenak (CoI)	University of Ljubljana, Dept. of Physics

OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
NIRCam Prime				
	1	NIRCam	NIRCam Imaging	(1) PlanckBulletMain
	2	NIRCam	NIRCam Imaging	(3) PlanckBulletSub
NIRSpec Prime				
	3	Plan 238 Primary	NIRSpec MultiObject Spectroscopy	(4) mockAPTprimary

ABSTRACT

The principal objective of this proposal is to study the nature of dark matter (DM) by using a merging galaxy cluster. Gravitational lensing provides a unique tool transforming these clusters into DM laboratories. The newly discovered Planck Bullet Cluster is an ideal and independent laboratory for detecting dark matter and distinguishing between cold dark matter (CDM) and other scenarios (e.g., self-interacting dark matter). With this proposal, the limit on the dark matter self-interaction cross-section will be measured (to $<0.5\text{cm}^2/\text{g}$ or better) to the extent that the possibility that dark matter self-interactions might explain the observed non-cuspy mass profiles can be fully excluded. To pin down the position of the DM component we require high-resolution, absolutely calibrated mass maps that will be compared to simulations. The combination of weak and strong lensing measurements is needed to attain this goal. This can only be achieved with the excellent resolving power and spectroscopic capabilities of JWST. We therefore request NIRCam and NIRSpec observations of the Planck Bullet Cluster. The combination of constraints from multiply lensed images (identified via color information and spectroscopy) and high-resolution weak lensing data will allow us to construct, self-consistently, the mass distribution from the very center to the outskirts. With this proposal we will further improve the selection of cluster members to unambiguously separate the baryonic and DM components. The proposal will enable investigations of the nature and properties of dark matter, and how it shapes galaxies and galaxy clusters and their evolution through cosmic time.

OBSERVING DESCRIPTION

The aim of these observations is to perform NIRCam imaging and NIRSpec spectroscopy of the Planck Bullet Cluster.

1. NIRCam Imaging with NIRISS on a parallel field.

Two NIRCam pointings, using a INTRAMODULEX dither pattern, are needed due to limited availability of ORIENTs and the geometry and subsequent imaging of the subcluster. Filters as follows SW: F090W, F115W, F150W, F200W and LW: F277W, F356W, F410M, F444W. The

NIRISS WFSS parallels use the F115W filter with both grisms and the F150W and F200W filters with only one grism. We understand observing in parallel with one grism in these 2 filters will lead to a higher fraction of targets affected by spectral contamination, however based on experience with NIRISS data such approach is manageable.

2. NIRSpec Multi-Object R=100 Spectroscopy on the cluster field with NIRCам imaging on a parallel field.

The NIRSpec multi-object spectroscopy will be targeted with a combination of NIRCам pre-imaging obtained in this program (1. above) and of sources from existing HST observations. High-priority targets are galaxies that are multiply imaged based on NIRCам imaging. The catalogs and MSA configurations in this APT file are therefore just placeholders to be changed when the NIRCам pre-imaging is obtained and processed. For NIRSpec Prime we will use NIRCам parallels with SW: F090W, F150W, F200W and LW: F356W, F410M, F444W.

3. ORIENT and schedule constraints:

We require at least 60 days between the NIRCам pre-imaging (both pointings) and NIRSpec observations to allow for data processing, analysis and target selection.

There is no ORIENT (V3PA) constraints, however we will need to slightly adjust the pointings, once the ORIENT is known for the observations. Observations are schedulable given all constraints.

Proposal 9594 - Targets - Planck Bullet: New Dark Matter Supercollider

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	PlanckBulletMain	RA: 11 57 57.3000 (179.4887500d) Dec: -10 46 1.00 (-10.76694d) Equinox: J2000	Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Clusters of Galaxies Description=[Galaxy mergers]				
	(3)	PlanckBulletSub	RA: 11 58 3.0000 (179.5125000d) Dec: -10 47 11.00 (-10.78639d) Equinox: J2000		
	Comments: Category=Clusters of Galaxies Description=[Galaxy mergers]				
	(4)	mockAPTprimary	RA: 11 58 0.3900 (179.5016250d) Dec: -10 46 32.90 (-10.77581d) Equinox: J2000		
	Comments: Description=[]				

Proposal 9594 - Observation 1 - Planck Bullet: New Dark Matter Supercollider

Observation	Proposal 9594, Observation 1: NIRCam										Wed May 13 20:00:19 GMT 2026		
	Diagnostic Status: Warning												
	Observing Template: NIRCam Imaging												
	Coordinated Parallel Template(s): NIRISS Wide Field Slitless Spectroscopy												
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(NIRCam (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous				
	(1)	PlanckBulletMain	RA: 11 57 57.3000 (179.4887500d) Dec: -10 46 1.00 (-10.76694d) Equinox: J2000			Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=Clusters of Galaxies Description=[Galaxy mergers]												
Template	NIRCam Imaging											NIRISS Wide Field Slitless Spectroscopy	
	Module: ALL												
	Subarray: FULL												
	Target Placement: Module A center (small extended source)												
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes	
	1	INTRAMODULEX		6				1		NIRCam Only		NO_DITHERING	
Spectral Elements	NIRCam Imaging		Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1		F150W	F277W	SHALLOW4	7	1	1	1	1	365.05		
	2		F150W	F277W	MEDIUM8	9	1	6	6	6	5669.015		
	3		F150W	F277W	SHALLOW4	7	1	1	1	1	365.05		
	4		F200W	F356W	SHALLOW4	7	1	1	1	1	365.05		
	5		F200W	F356W	MEDIUM8	9	1	6	6	6	5669.015		
	6		F200W	F356W	SHALLOW4	7	1	1	1	1	365.05		
	7		F115W	F444W	SHALLOW4	7	1	1	1	1	365.05		
	8		F115W	F444W	MEDIUM8	9	1	6	6	6	5669.015		
	9		F115W	F444W	SHALLOW4	7	1	1	1	1	365.05		
	10		F090W	F410M	SHALLOW4	7	1	1	1	1	365.05		
	11		F090W	F410M	MEDIUM8	9	1	6	6	6	5669.015		
	12		F090W	F410M	SHALLOW4	7	1	1	1	1	365.05		

Proposal 9594 - Observation 1 - Planck Bullet: New Dark Matter Supercollider

	NIRISS Wide Field Slitless Spectroscopy	Exposure Type	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp	Two Extra Dithers	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
Spectral Elements	1	DIRECT	F115W		NIS	8	1	NO	1	1	354.313	
	2	GRISM	F115W	GR150C	NIS	21	1	6	6	5475.753		3
	DIRECT	F115W		NIS	8	1	NO	1	1	354.313		4
	DIRECT	F115W		NIS	8	1	NO	1	1	354.313		5
	GRISM	F115W	GR150R	NIS	21	1	6	6	5475.753		6	DIRECT
	F115W		NIS	8	1	NO	1	1	354.313		7	DIRECT
	F150W		NIS	8	1	NO	1	1	354.313		8	GRISM
	F150W	GR150C	NIS	21	1	6	6	5475.753		9	DIRECT	F150W
		NIS	8	1	NO	1	1	354.313		10	DIRECT	F200W
		NIS	8	1	NO	1	1	354.313		11	GRISM	F200W
	GR150C	NIS	21	1	6	6	5475.753		12	DIRECT	F200W	
Special Requirements	Aperture PA Range 97.86565232 to 117.86565232 Degrees (V3 98.0 to 118.0) Aperture PA Range 283.86565232 to 313.86565232 Degrees (V3 284.0 to 314.0) No Parallel Attachments Fiducial Point Override NRCAS_FULL 2 After 1 by <None specified> to 10 Days 3 After 1 by 60.0 Days to <None specified> Same Aperture PA 1, 2											

Proposal 9594 - Observation 2 - Planck Bullet: New Dark Matter Supercollider

Observation	Proposal 9594, Observation 2: NIRCam										Wed May 13 20:00:19 GMT 2026		
	Diagnostic Status: Warning												
	Observing Template: NIRCam Imaging												
	Coordinated Parallel Template(s): NIRISS Wide Field Slitless Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
	(NIRCam (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(3)	PlanckBulletSub	RA: 11 58 3.0000 (179.5125000d) Dec: -10 47 11.00 (-10.78639d) Equinox: J2000										
	Comments: Category=Clusters of Galaxies Description=[Galaxy mergers]												
Template	NIRCam Imaging											NIRISS Wide Field Slitless Spectroscopy	
	Module: ALL												
	Subarray: FULL												
	Target Placement: Module A center (small extended source)												
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes	
	1	INTRAMODULEX		6				1		NIRCam Only		NO_DITHERING	
Spectral Elements	NIRCam Imaging		Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1		F150W	F277W	SHALLOW4	7	1		1	1	365.05		
	2		F150W	F277W	MEDIUM8	9	1		6	6	5669.015		
	3		F150W	F277W	SHALLOW4	7	1		1	1	365.05		
	4		F200W	F356W	SHALLOW4	7	1		1	1	365.05		
	5		F200W	F356W	MEDIUM8	9	1		6	6	5669.015		
	6		F200W	F356W	SHALLOW4	7	1		1	1	365.05		
	7		F115W	F444W	SHALLOW4	7	1		1	1	365.05		
	8		F115W	F444W	MEDIUM8	9	1		6	6	5669.015		
	9		F115W	F444W	SHALLOW4	7	1		1	1	365.05		
	10		F090W	F410M	SHALLOW4	7	1		1	1	365.05		
	11		F090W	F410M	MEDIUM8	9	1		6	6	5669.015		
	12		F090W	F410M	SHALLOW4	7	1		1	1	365.05		

Proposal 9594 - Observation 2 - Planck Bullet: New Dark Matter Supercollider

	NIRISS Wide Field Slitless Spectroscopy	Exposure Type	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp	Two Extra Dithers	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
Spectral Elements	1	DIRECT	F115W		NIS	8	1	NO	1	1	354.313	
	2	GRISM	F115W	GR150C	NIS	21	1	6	6	5475.753		3
	DIRECT	F115W		NIS	8	1	NO	1	1	354.313		4
	DIRECT	F115W		NIS	8	1	NO	1	1	354.313		5
	GRISM	F115W	GR150R	NIS	21	1	6	6	5475.753		6	DIRECT
	F115W		NIS	8	1	NO	1	1	354.313		7	DIRECT
	F150W		NIS	8	1	NO	1	1	354.313		8	GRISM
	F150W	GR150C	NIS	21	1	6	6	5475.753		9	DIRECT	F150W
		NIS	8	1	NO	1	1	354.313		10	DIRECT	F200W
		NIS	8	1	NO	1	1	354.313		11	GRISM	F200W
	GR150C	NIS	21	1	6	6	5475.753		12	DIRECT	F200W	
Special Requirements	Aperture PA Range 98.41565232 to 117.86565232 Degrees (V3 98.55 to 118.0) Aperture PA Range 283.86565232 to 313.86565232 Degrees (V3 284.0 to 314.0) No Parallel Attachments Fiducial Point Override NRCAS_FULL 2 After 1 by <None specified> to 10 Days Same Aperture PA 1, 2											

Proposal 9594 - Observation 3 - Planck Bullet: New Dark Matter Supercollider

Observation	Proposal 9594, Observation 3: Plan 238 Primary											Wed May 13 20:00:19 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec MultiObject Spectroscopy											
	Coordinated Parallel Template(s): NIRCcam Imaging											
Diagnostics	(Plan 238 Primary (Obs 3)) Warning (Form): Config c1 (#1) has 1 primary slits affected by failed closed shutters.											
	(Plan 238 Primary (Obs 3)) Warning (Form): Config c2 (#2) has 1 primary slits affected by failed closed shutters.											
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 3:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Visit 3:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	mockAPTprimary	RA: 11 58 0.3900 (179.5016250d) Dec: -10 46 32.90 (-10.77581d) Equinox: J2000									
	Comments: Description=[]											
Acquisition	NIRSpec MultiObject Spectroscopy	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
	3		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788		
Template	NIRSpec MultiObject Spectroscopy						NIRCcam Imaging					
	TA Method: MSATA						Module: ALL					
	HFF Readout Mode: false						Subarray: FULL					
	Obtain Confirmation Images: No											
	Science Aperture: MSA Center											
	Primary Candidate List: mockAPTprimary (63 sources)											
	Filler Candidate List: mockAPTprimary (63 sources)											
	Spectral Overlap Map: jwst-nirspec-prism											
	Spectral Overlap Threshold: 1.5											
Reference Stars												
Dithers	#	Dither Type										
	1	NONE										

Proposal 9594 - Observation 3 - Planck Bullet: New Dark Matter Supercollider

Spectral Elements	NIRSpec MultiObject Spectroscopy	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
	1	1 (PRISM/CLEAR)	c1	3 Shutter Slitlet	179.48946562499 998 Degrees - 10.790633055555 531 Degrees	238.00230803110 8			3	3	2888.6
	2	1 (PRISM/CLEAR)	c2	3 Shutter Slitlet	179.514106875 Degrees - 10.764370833333 317 Degrees	237.99763874633 626			3	3	2888.6
	3	1 (PRISM/CLEAR)	c3	3 Shutter Slitlet	179.49426195833 334 Degrees - 10.787688888888 908 Degrees	238.00140361605 59			3	3	2888.6
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F277W	MEDIUM8	9	1	3	3	2834.507		
	2	F150W	F356W	MEDIUM8	9	1	3	3	2834.507		
	3	F200W	F444W	MEDIUM8	9	1	3	3	2834.507		
Special Requirements	Sequence Visits within 53.0 Days Visits Same PA No Parallel Attachments MSA Planned Aperture PA 238.0000 to 238.0000 Degrees (V3 99.4254303 to 99.4254303) 3 After 1 by 60.0 Days to <None specified>										